Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

< reasoning process here >

To find the de Broglie wavelength of an electron with a kinetic energy of 1 eV, we use the de Broglie wavelength formula:

Î» = h / p

where:
- h is Planck's constant = 6.626 ÃĹ 10âģ»Â³âģ´ JÂ·s (or 6.626 ÃĹ 10âģ»Â²âģ· ergÂ·s)
- p is the momentum of the electron

For non-relativistic electrons, kinetic energy (K.E.) is related to momentum by:

K.E. = (1/2)mvÂ² = pÂ² / (2m)

So, solving for momentum p:

p = âĪļ(2mK.E.)

Thus,

Î» = h / âĪļ(2mK.E.)

We can plug in values to compute the wavelength.

Step 1: Convert KE to joules.

1 eV = 1.602 ÃĹ 10âģ»Â¹âģ¹ J

So, K.E. = 1.602 ÃĹ 10âģ»Â¹âģ¹ J

Step 2: Mass of electron m = 9.109 ÃĹ 10âģ»Â³Â¹ kg

Step 3: Compute momentum:

p = âĪļ(2 ÃĹ 9.109e-31 ÃĹ 1.602e-19)
= âĪļ(2 ÃĹ 9.109 ÃĹ 1.602 ÃĹ 10âģ»âģµâģ°)
âīĪ âĪļ(2.918 ÃĹ 10âģ»âģµâģ°)
âīĪ 1.708 ÃĹ 10âģ»Â²âģµ kgÂ·m/s

Step 4: Compute Î» = h / p

h = 6.626 ÃĹ 10âģ»Â³âģ´ JÂ·s

Î» = 6.626e-34 / 1.708e-25 âīĪ 3.88 ÃĹ 10âģ»âģ¹ m

Convert to cm: 3.88 ÃĹ 10âģ»âģ¹ m = 3.88 ÃĹ 10âģ»âģ· cm

Now, check the options:

A. 2.48 ÃĹ 10âģ»âģ¸ cm âĨĴ too small
B. 5.92 ÃĹ 10âģ· cm/sec âĨĴ units are speed, not wavelength âĨĴ invalid
C. 8.66 ÃĹ 10âģ»âģ¶ cm âĨĴ too large
D. 3.14 ÃĹ 10âģ»âģµ cm âĨĴ too large
E. 9.81 ÃĹ 10âģ»âģ´ cm âĨĴ too large
F. 5.27 ÃĹ 10âģ»âģ¹ m âĨĴ 5.27 ÃĹ 10âģ»Â¹Â¹ cm âĨĴ too small
G. 1.23 ÃĹ 10âģ»âģ· cm âĨĴ = 1.23 ÃĹ 10âģ»âģ¹ m âĨĴ too small
H. 4.14 ÃĹ 10âģ»Â¹âģ° m âĨĴ = 4.14 ÃĹ 10âģ»Â¹Â² cm âĨĴ too small
I. 6.625 ÃĹ 10âģ»Â²âģ· ergÂ·s âĨĴ units are h, not wavelength
J. 0.512 ÃĹ 10âģ¶ eV âĨĴ energy, not wavelength

But our computed value is approximately 3.88 ÃĹ 10âģ»âģ¹ m = 3.88 ÃĹ 10âģ»âģ· cm

Wait âĢĶ let's check if we made a unit error.

We found: Î» âīĪ 3.88 ÃĹ 10âģ»âģ¹ m = 3.88 ÃĹ 10âģ»âģ· cm

But among the options:

F is 5.27 ÃĹ 10âģ»âģ¹ m âĨĴ 5